

Beginner tips to write scientific documents

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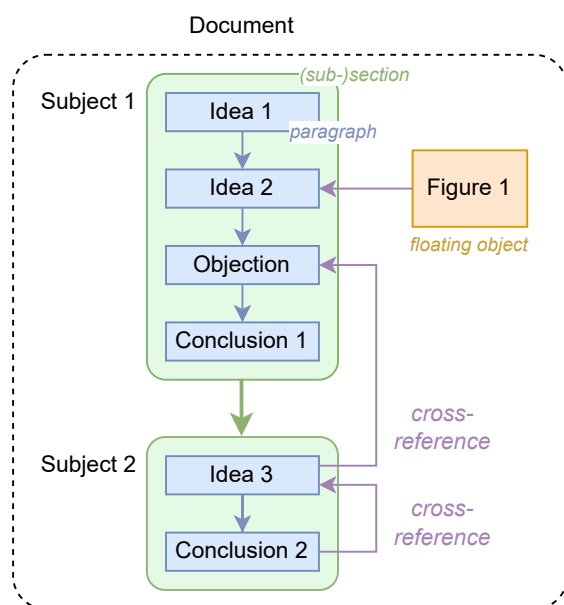


Figure 1: Example of a document seen as an interconnection of blocks. Blue blocks are paragraphs; green blocks are (sub-)sections; orange blocks are floats.

We denote recommended options by ✓ and discouraged options by ✗.

1 High-level tips

Structure Before writing the document, make an outline of *all* sections. Then, optionally write a short sentence for each paragraph.

Motivation A text can be seen as an *ordered collection of interconnected blocks*, the blocks being paragraphs or sections; see Figure 1. The outline helps you clarify the content of each block and their relations, which will result in a clearer and better organized document.

Consistency In principle, you can ignore any tip written here, but if you do so, ponder *why* you are doing that, and maintain *consistency*. If consistency is lacking, deliberate choices appear as mistakes.

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2 Tips for writing

2.1 Language

American/British English Commit to using either American or British English and do not mix the two.

Contractions Do not use contractions in formal documents.
Examples ✗ “We’ve shown”. ✓ “We have shown”.

2.2 Style

Specific nouns Use uppercase for proper nouns. Use italics the first time you introduce a specific concept. Use quotes to quote text or report titles.

Examples

- “The *input flow* is the amount of flow entering the machine. Hence, input flow is an important quantity”.
- “This topic is discussed in “On Convergence” by John Smith”.
- “Deep learning became a popular topic in computer science” or “Deep Learning became a popular topic in Computer Science”.

Note There is no strong consensus on whether to capitalize names of disciplines and algorithms. We advise *not* to capitalize them, as it can become difficult to establish which is important enough to deserve the treatment.

Acronyms Spell out acronyms the first time they are used.

Examples ✗ “MPC has many advantages”. ✓ “Model predictive control (MPC) has many advantages”.

Exception Well-known acronyms, in general or in the relevant context (e.g., USA or EU, in general; PID, in control theory).

Emphasis Communicate emphasis with italics, not bold.

Examples ✗ “The phenomenon is **internal** to the machine”. ✓ “The phenomenon is *internal* to the machine”.

Motivation Bold can be distracting.

Dashes There are three types of dashes, the *hyphen* (“-”), the *en dash* (“—”) and the *em dash* (“—”). Use hyphens to link words. Use en dashes to denote ranges of numbers. Use em dashes to write parenthetical clauses.

Examples “Editor-in-chief”. “Pages 23–25”. “An important—yet sometimes neglected—concept”.

How to In $\text{\LaTeX}$, to type the hyphen, the en dash, and the em dash, use -, --, and ---, respectively.

2.3 Structure

Paragraphs and new lines New lines are obtained typing `\` in $\text{\LaTeX}$, and Shift+Enter in Word. Paragraphs are obtained typing a blank line or `\par` in $\text{\LaTeX}$, and Enter in Word. Avoid using new lines.

Motivation The sub-units of sections are paragraphs not “portions of text separated by new lines”. Using both makes the structure of the document unclear.

Inline enumeration Optionally, denote inline lists by using (i), (ii), (iii), etc.

Example “There are three facts to underline. (i) The event generates wealth, (ii) it can be repeated, and (iii) it is uncommon.”

Motivation Can be used to save space and/or avoid breaking the flow of the text. Using (1), (2), etc. would conflict with cross-referencing of equations.

Cross-references A cross-reference is a reference to another part of the text. Use them as much as possible.

Examples of notation For sections, “Section 3.2” or “§ 3.2”. For figures, “Figure 6”. For tables, “Table 6”. for equations, “(5)” For math environments “Definition 4” or “Theorem 4” and so on.

Examples of use ✗ “In the figure below, we explain the phenomenon described in the previous section”. ✓ “In Figure 3.2, we explain the phenomenon described in Section 2.1 (see (3) in Theorem 2)”.

Motivation Cross-referencing helps the author ensure the content is correct and helps the reader understand it.

How to In $\text{\LaTeX}$, use `\label`, `\ref` and `\eqref`.

Cross-referencing of floats In $\text{\LaTeX}$, floats or (floating objects) are items such as figures and tables whose exact position in the text is flexible and decided by the compiler. Insert at least one cross-reference for each float; see Figure 1.

Motivation If a float is never referenced in the text, the reader might not understand what part it is related to.

2.4 Other parts of text

Captions In a caption (of a figure or a table), insert enough information to understand what is portrayed.

Examples ✗ “Numerical simulations”. ✓ “20 repetitions of the numerical simulations described in Section 3.2. The black solid line portrays current; the red dashed line represents voltage”.

References Use references to justify statements concerning the state of the art.

Examples ✗ “Many solutions were proposed to control engines”. ✓ “Many solutions were proposed to control engines (e.g., [3], [4], [5])”.

3 Mathematics

3.1 General concepts

Math in proper environment Enclose mathematical terms in math environments.

Examples ✓ J is the objective function. ✗ J is the objective function.

How to In $\text{\LaTeX}$, for inline equations, use $\$ \dots \$$ or $\text{\textbackslash}(\dots\text{\textbackslash})$.

Punctuation and math When using punctuation, treat math symbols and whole equations as if they were nouns or phrases.

Examples

- ✗ “The objective function is important $J = a + b$ can be computed using a computer.”
- ✓ “The important objective function J is given as $J = a + b$, which can be computed using a computer”.
- ✓ Even though it holds that $a > b$, it also holds that $b = c$.

Alignment of multiple equations When using multiple equations, align them at the equation/inequality sign, using proper environments (e.g., in $\text{\LaTeX}$, `align`).

Examples

- $$\begin{aligned} \text{✗ } a &= b + c + d, \\ e &= f. \end{aligned}$$
- $$\begin{aligned} \text{✓ } a &= b + c + d, \\ e &= f. \end{aligned}$$

Measurement units Write measurement units in upright font, separated from the number by a small space (in $\text{\LaTeX}$, `\,`).

Examples ✗ “The speed is $5m/s$.” ✓ “The speed is $5\,m/s$ ”.

3.2 Definitions

Definitions Define variables and functions (preferably) before they are used, or immediately afterwards.

Symbol for definition Denote explicitly when a relation is a definition.

Examples (assuming a is defined as being equal to b^2), ✓ “ $a \triangleq b^2$ ”. ✓ “ $a := b^2$ ”. ✗ “ $a = b^2$ ”.

How to In $\text{\LaTeX}$, “ $\triangleq$ ” is properly typeset by typing `\coloneqq` (using the `mathtools` package); “ $\trianglelefteq$ ” is obtained by typing `\trianglelefteq`.

Domains and codomains When defining a variable, specify the set it belongs to. When defining a function, specify its domain and codomain.

Examples

- ✗ “Let v be the speed and f be the absolute value function.”
- ✓ “Let $v \in \mathbb{R}$ be the speed and $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be the absolute value function.”

Top-down and bottom-up presentations When presenting complex relations, use a *top-down* or *bottom-up* approach.

Examples

- (Top-down) “ a is given by $a = b + c$, where b is $\dots$, and c is $\dots$.”
- (Bottom-up) “Let b be $\dots$, and c be $\dots$. Then, a is given by $a = b + c$.”

Usage Typically, the top-down form is quicker and works better for simpler relations; the bottom-up form is less immediate but more suited for involved relations.

3.3 Notation for symbols

Symbols conventions Whenever possible, be consistent with the use of symbols and of groups of symbols.

Examples $\hat{}$ denote estimated quantities. Upper-case letters denote matrices. Bold letters denote random variables.

Note There does not exist a comprehensive one-size-fits-all set of rules. Choose those that best serve your specific case.

Known sets of numbers Use $\mathbb{N}$ for the set of natural numbers, $\mathbb{Z}$ for real numbers, $\mathbb{Q}$ for rational numbers, $\mathbb{R}$ for real numbers, and $\mathbb{C}$ for complex numbers. For irrational numbers, state what symbol you are using (e.g., $\mathbb{Q}^+$, $\mathbb{I}$). Use subscripts such as $_{>0}$ to denote whether the sets contain zero and/or positive/negative numbers (e.g., $\mathbb{N}_{>0}$, $\mathbb{R}_{\geq 0}$). Avoid ambiguous symbols such as $\mathbb{R}^+$.
How to In $\LaTeX$, use `\mathbb{b}` (from the `amsfonts` package).

Operators in upright Typeset math operators in upright font.

Examples (denoting the rank of matrix by “rk”), ✓ “rk(A)”. ✗ “rk(A)”.

Motivation In rare cases, operators in italics might be misunderstood as products of variables.

The d in dx In integrals and derivatives, typeset the symbol d in upright font.

Examples ✗ “ $\int x^2 dx$ ”. ✓ “ $\int x^2 dx$ ”. ✗ “ dx/dt ”. ✓ “ dx/dt ”.

Motivation d denotes specific operations, it is not a variable.

3.4 Subscripts and superscripts

Subscripts and superscripts Write subscripts and superscripts in upright font, except when they refer to specific mathematical objects.

Examples Let $a \in \mathbb{R}$. An error pertaining to variable a could be denoted by e_a . Differently, an absolute error (unrelated to a) could be denoted by e_a .

Note For clarity, try however to avoid situations where the same sub/super-script appears both in italics and in uprights, with different meanings.

Sub/super-scripts with multiple elements In sub/super-scripts with multiple entries, separate numerical entries with commas.

Examples ✓ A_{ij} , $A_{i,j}$, $A_{2,2}$. ✗ A_{22} .

3.5 Brackets

Sets and tuples Denote (unordered) sets through curly braces and tuples (i.e., ordered sets) through round parentheses.

Examples $\{1, 2, \dots, 5\}$ is a set; $(1, 2, 5, 3, 2)$ is a tuple.

Intervals Use $[a, b]$, $]a, b]$, $[a, b[$, $]a, b[$ for continuous intervals in $\mathbb{R}$ and avoid round parentheses. Use $\{a, \dots, b\}$ for intervals in $\mathbb{Z}$.

Motivation If (a, b) were used for an interval, it could be mistaken for a pair (a 2-tuple).

Vectors and matrices Denote vectors and matrices through square (or round) brackets, with elements separated by spaces.

Examples $v = [v_1 \ v_2 \ v_3]$.

Motivation If commas were used, vectors might be mistaken for intervals or tuples.

3.6 Figures

File format Figures should be saved in a vector format (e.g., .pdf, .svg, .eps) rather than a raster one (e.g., .jpg, .png, .bmp).

Motivation Vector formats are scalable and do not lose quality when zoomed in.

Source material Store a copy of the source material for each figure (e.g., in Matlab, .fig files; in Python, .pkl or .pickle).

Motivation This allows editing of the figures without having to recreate them from scratch.

Figure size To avoid distortions, in the document embed figures at their original size, without scaling. To allow this, save figures with a width that is at most equal to the text width.

How to ($\LaTeX$)

1. Find out the values of `\columnwidth` (for single-column figures) and `\textwidth` (for double-column figures).¹
2. Save figures with a width of at most `\columnwidth` or `\textwidth`.
3. Embed figures without scaling or with a maximum width.²

Text in figures If possible, in figures saved in a vector format, ensure text is saved as proper text and not as *paths*.

Motivation Text saved as path cannot be easily edited.

Fonts When possible, the font face and size should be the same as those used in the main text.

Example Many academic journals use Times New Roman at 10 points, whereas some publishers use proprietary fonts.

¹E.g., using `\usepackage{layouts}` and, e.g., `\printinunitsof{cm}{\prntlen{\columnwidth}}`.

²To embed without scaling, use `\includegraphics{file_name}`. To embed with a maximum width, use `\usepackage[export]{adjustbox}`, then, e.g., `\includegraphics[max width=\columnwidth]{file_name}`.